

Disordered $J - J'$ Model in a Triangular Lattice; A Mean-field Approximation

The disordered next-nearest neighbour Heisenberg model is described by the Hamiltonian,

$$H = \sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \sum_{\langle\langle i,j \rangle\rangle} J'_{ij} \mathbf{S}_i \cdot \mathbf{S}_j \quad (1)$$

, where both the nearest and the next nearest coupling, J and J' are drawn from random positive definite distributions. The distribution is taken to be positive definite so that the system remains anti-ferromagnetic. To find a mean-field description of this model it is useful to trade the spin operators for their slave fermion representation which makes way for mean-field decoupling in various channels of excitation. The generalised $SU(N)$ model can be exactly solved in this paradigm. However, such a scheme fails to capture any emergent magnetic order of the system and therefore remains ill-suited to describe the magnetic phases of the disordered system. An Ising type mean-field description which approximates the model in terms of local moments and local fields remain an avenue to describe the magnetic phases. In the following we concentrate on a hybrid scheme which involves both these types of mean-field decoupling by introducing a cosmetic parameter.

I. THE MEAN-FIELD DECOUPLING

The description of the mean-field decoupling are the same for both the nearest and the next-nearest neighbour part of the hamiltonian. In the slave fermion representation for the $SU(2)$ model, the spin operators can be expressed as $\mathbf{S}_i = f_{i\alpha}^\dagger \boldsymbol{\sigma}_{\alpha\beta} f_{i\beta}$ with the constraint $f_{i\alpha}^\dagger f_{i\alpha} = 1$. In the notation that follows, only the fermion flavours (greek letters) are summed over if repeated and the sum over the lattice coordinates are explicitly mentioned. An additional parameter x allows us to include both the decoupling in the fermion-fermion channel and also on the basis of the local moments. The mean-field hamiltonian following this scheme is obtained to be,

$$H_{\text{mf}} = - \sum_{\langle i,j \rangle} \left(\chi_{i,j}^{N*} f_{i\alpha}^\dagger f_{j\alpha} + \text{h.c.} \right) - \sum_{\langle\langle i,j \rangle\rangle} \left(\chi_{ij}^{NN*} f_{i\alpha}^\dagger f_{j\alpha} + \text{h.c.} \right) \\ + \frac{1}{2} \sum_i \mathbf{B}_i \cdot f_{i\alpha}^\dagger \boldsymbol{\sigma}_{\alpha\beta} f_{i\beta} + \sum_i \lambda_i \left(f_{i\alpha}^\dagger f_{i\alpha} - 1 \right) \quad (2)$$

with the following set of equation determining the mean-field parameters.

$$\begin{aligned}
\langle f_{i\alpha}^\dagger f_{i\alpha} \rangle &= 1 \\
\chi_{ij}^N &= (1-x) J_{ij} \langle f_{i\alpha}^\dagger f_{j\alpha} \rangle, \quad i,j \in N \\
\chi_{ij}^N &= (1-y) J_{ij} \langle f_{i\alpha}^\dagger f_{j\alpha} \rangle, \quad i,j \in NN \\
B_i &= \frac{1}{2} \left[x \sum_{j \in N} J_{ij} \langle f_{j\alpha}^\dagger \sigma_{\alpha\beta} f_{j\beta} \rangle + y \sum_{j \in NN} J_{ij} \langle f_{j\alpha}^\dagger \sigma_{\alpha\beta} f_{j\beta} \rangle \right]
\end{aligned} \tag{3}$$

It is possible to solve this system of equation in real space using exact diagonalisation and iterative convergence procedure.

II. HUBBARD-STRATANOVICH TRANSFORMATION: FORMAL BOSONISATION

Like the mean-field decoupling expressed describe it is possible to trade an interacting Fermionic problem for a problem with additional bosons coupled with the fermions. In its formulation, this constitutes an exact description of the original problem and leads way for further exploration using different applicable techniques. One of the formal way to achieve this is to perform the Hubbard-Stratanovich transformation, which, in its content is very much an integral identity for gaussian form.

To illustrate the method we consider a generic interacting problem involving fermions ψ_i in a lattice. Let the Hamiltonian be given by,

$$H = \sum_{ij} t_{ij} \psi_i^\dagger \psi_j + \sum_{ijkl} U_{ijkl} \psi_i^\dagger \psi_j^\dagger \psi_k \psi_l \tag{4}$$

. We are interested in the partition function of the system given by, $Z = \text{Tr} e^{-\beta(H-\mu N)}$ with the number operator, $N = \sum_i \psi_i^\dagger \psi_i$. Now, to get to the functional integral formalism we introduce the time-slicing scheme. In the stastical partition function there is no time ordered integral, however, the exponent can still be written down as sum of infinite number of exponents, $\text{Tr} e^{-\beta(H-\mu N)} = \text{Tr} e^{-\sum_{t=1}^T \epsilon(H-\mu N)}$ such that, $T\epsilon = \beta$. This construction allows us to express the exponential in terms of multiplication of multiple time-ordered exponentials upto some trotter error and introduce resolutions of identities in terms of the time-stamped Grassmann coherent states.

A. Grassmann Coherent States

Grassmann numbers anti-commute, such that for two Grassmann numbers, $\alpha\beta + \beta\alpha = 0$. As a consequence of this anti-commutation they are also associated with the following properties,

$$\begin{aligned}\int d\alpha 1 &= 0 \\ \int d\alpha \alpha &= 1 \\ \frac{\partial}{\partial \alpha} \alpha &= 1 \\ \frac{\partial}{\partial \alpha} 1 &= 0\end{aligned}\tag{5}$$

It is noteworthy that the integrals are defined the same way as the differentials. Unlike real or complex numbers these can not readily be interpreted in terms of surface areas or infinitesimal increments. However these relations are self-consistent and they uniquely define the Grassmann calculus. At any stages of a calculation involving physical fermionic observables these relations can act as mnemonics and simplify the calculational approach.

Fermionic operators anti-commute so Grassmann numbers become ubiquitous for the expression of Grassman coherent states. The definition of the fermionic coherent states is the following,

$$|\psi\rangle = e^{-\sum_i \Psi_i \psi_i^\dagger} |0\rangle = \prod_i \left(1 - \Psi_i \psi_i^\dagger\right) |0\rangle\tag{6}$$

, where Ψ_i are Grassmann variables and $|0\rangle$ is the multi-particle vacuum. The expression of the exponential in terms of multiplication here is made possible by the fact that the objects in the sum of the exponent commute between themselves and the Taylor expansion of Grassmann numbers have no non-linear non-zero terms.

It is straightforward to verify that, $\psi_i |\psi\rangle = \Psi_i |\psi\rangle$. Similarly for the adjoint of the coherent state we need a different flavour of Grassmann numbers, Ψ_i^* and the expression, $\langle\psi| = \langle 0| e^{-\sum_i \phi_i \Psi_i^*}$ fulfills the second criteria as a left coherent state, $\langle\psi| \psi_i^\dagger = \langle\psi| \Psi_i^*$. It can also be shown that, $\psi_i^\dagger |\psi\rangle = -\frac{\partial}{\partial \Psi_i} |\psi\rangle$ and $\langle\psi| \psi_i = +\frac{\partial}{\partial \Psi_i^*} \langle\psi|$. It is worth noting that the different flavours of Grassmann numbers Ψ_i and Ψ_i^* are not related in the usual sense of complex conjugation. The Grassmann coherent states are over-complete, $\langle\psi|\psi\rangle = e^{\sum_i \Psi_i^* \Psi_i}$. This leads to the closure relation,

$$\int \prod_i d\Psi_i^* d\Psi_i e^{-\sum_i \Psi_i^* \Psi_i} |\psi\rangle \langle\psi| = 1\tag{7}$$

Although the prefactor is apprant from the expression of over-completeness of the coherent states it is useful to see how the above expression acts as an identity operator in the Fock space of the many body system. We take the operator, $A = \int \prod_i d\Psi_i^* d\Psi_i e^{-\sum_i \Psi_i^* \Psi_i} |\psi\rangle\langle\psi|$ and find that the matrix element of this operator between two arbitrary vectors in the Fock space is given by,

$$\begin{aligned}
& \langle\beta_1, \beta_2, \dots, \beta_N | A | \alpha_1, \alpha_2, \dots, \alpha_N \rangle \\
&= \int \prod_i d\Psi_i^* d\Psi_i e^{-\sum_i \Psi_i^* \Psi_i} \langle\beta_1, \beta_2, \dots, \beta_N | \psi\rangle \langle\psi | \alpha_1, \alpha_2, \dots, \alpha_N \rangle \\
&= \int \prod_i d\Psi_i^* d\Psi_i \prod_i (1 - \Psi_i^* \Psi_i) \langle 0 | \psi_{\beta_1}, \psi_{\beta_2} \dots \psi_{\beta_N} | \psi \rangle \langle \psi | \psi_{\alpha_1}^\dagger, \psi_{\alpha_2}^\dagger, \dots, \psi_{\alpha_N}^\dagger | 0 \rangle \\
&= \int \prod_i d\Psi_i^* d\Psi_i (1 - \Psi_i^* \Psi_i) \Psi_{\beta_1} \Psi_{\beta_2} \dots \Psi_{\beta_N} \Psi_{\alpha_1}^* \Psi_{\alpha_2}^* \dots \Psi_{\alpha_N}^* \\
&= \int \prod_i d\Psi_i^* d\Psi_i (1 - \Psi_i^* \Psi_i) (-1)^P \delta_{\beta, \alpha} \Psi_{\beta_1} \Psi_{\beta_2} \dots \Psi_{\beta_N} \Psi_{\beta_1}^* \Psi_{\beta_2}^* \dots \Psi_{\beta_N}^* \\
&\text{for } P \{ \alpha_1, \alpha_2, \dots, \alpha_N \} = \{ \beta_1, \beta_2, \dots, \beta_N \} \\
&= \int \prod_i d\Psi_i^* d\Psi_i (1 - \Psi_i^* \Psi_i) (-1)^P \delta_{\beta, \alpha} \Psi_{\beta_1} \Psi_{\beta_1}^* \Psi_{\beta_2} \Psi_{\beta_2}^* \dots \Psi_{\beta_N} \Psi_{\beta_N}^* \\
&= (-1)^P \delta_{\beta, \alpha} = \langle\beta_1, \beta_2, \dots, \beta_N | \alpha_1, \alpha_2, \dots, \alpha_N \rangle
\end{aligned} \tag{8}$$

where the δ function determines whether the Fock index vector β is same as α upto a permutation P . The exponential factor $e^{-\sum_i \Psi_i^* \Psi_i}$ is crucial for the formalism because without it the Grassmann integral, $\int d\Psi_i^* d\Psi_i (1 - \Psi_i^* \Psi_i) \Psi_{\beta_1} \Psi_{\beta_1}^*$ would yield to 0 if $i \neq \beta_1$. So, the formalism for Grassmann integral and the Fock-space identity, $\int \prod_i d\Psi_i^* d\Psi_i e^{-\sum_i \Psi_i^* \Psi_i} |\psi\rangle\langle\psi| = 1$, remains internally consistent.

B. The Functional Integral Formalism and The Hubbard-Stratanovich Transformation

With the algebra for Fermionic-coherent state set up we can return to our original problem, i.e. to describe the partition function of the interacting Fermion model in a lattice in terms of functional integrals. The expression for the partition function is, $Z = \sum_n \langle n | e^{-\beta(H - \mu N)} | n \rangle$, where $|n\rangle$ are the Fock-space eigenstates of the Many-body hamiltonian. Since the overlap between the Fock-space eigenstates and the coherent states, $\langle n | \psi \rangle$ contain Grassmann numbers, we have the following identity, $\langle n | \psi \rangle \langle \psi | n \rangle = \langle -\psi | n \rangle \langle n | \psi \rangle$. This relation is usefull to exchange the trace over the Fock-space states with traces over Fermionic coherent

states.

$$\begin{aligned}
Z &= \sum_n \langle \mathbf{n} | e^{-\beta(H-\mu N)} | \mathbf{n} \rangle \\
&= \sum_n \left(\int \prod_i d\Psi_i^* d\Psi_i e^{-\sum_i \Psi_i^* \Psi_i} \right)^2 \langle \mathbf{n} | -\psi \rangle \langle -\psi | e^{-\beta(H-\mu N)} | \psi \rangle \langle \psi | \mathbf{n} \rangle \\
&= \int \prod_i d\Psi_i^* d\Psi_i e^{-\sum_i \Psi_i^* \Psi_i} \langle -\psi | e^{-\beta(H-\mu N)} | \psi \rangle
\end{aligned} \tag{9}$$

Since our hamiltonian is expressed in terms of the creation and annihilation operators, the 'time' slicing and subsequent insertion of the Fermionic coherent states re-expresses the partion function in terms of Functional integrals over the field of time stamped Grassmann numbers. Now, with the absence of any time integral in the exponent the formal way to introduce time slicing, $N\epsilon = \beta$ would require a symbolic label for the Fermionic coherent states and their respective Grassmann fields.

$$\begin{aligned}
Z &= \int_{\psi(N)=-\psi(1)} \prod_{i,t=1}^{t=N-1} d\Psi_i^*(t) d\Psi_i(t) e^{-\sum_{i,t} \Psi_i^*(t) \Psi_i(t)} \\
&\quad \langle \psi(N) | : e^{-\epsilon(H-\mu N)} : | \psi(N-1) \rangle \dots \langle \psi(2) | : e^{-\epsilon(H-\mu N)} : | \psi(1) \rangle
\end{aligned} \tag{10}$$

. Here, $::$ deontes the normal ordering of the operators (i.e. the creation operators are all to the left of the annihilation operators) and the Trotter error arising from truncating the Baker-Campbell-Hausdroff formula upto the linear term (which is $\mathcal{O}(\epsilon^2)$) is ignored. The matrix-element of the exponentiated Hamiltonian (4) is given by,

$$\begin{aligned}
\langle \psi(t) | e^{-\epsilon(H-\mu N)} | \psi(t-1) \rangle &= \langle \psi(t) | 1 - \epsilon \left(\sum_{ij} (t_{ij} - \mu \delta_{ij}) \psi_i^\dagger \psi_j + \sum_{ijkl} U_{ijkl} \psi_i^\dagger \psi_j^\dagger \psi_k \psi_l \right) | \psi(t-1) \rangle \\
&= 1 + \sum_i \Psi_i^*(t) \Psi_i(t-1) \\
&\quad - \epsilon \left(\sum_{ij} (t_{ij} - \mu \delta_{ij}) \Psi_i^*(t) \Psi_j(t-1) + \sum_{ijkl} U_{ijkl} \Psi_i^*(t) \Psi_j^*(t) \Psi_k(t-1) \Psi_l(t-1) \right) \\
&= e^{\sum_i \Psi_i^*(t) \Psi_i(t-1) - \epsilon H(\Psi^*(t), \Psi(t-1))}
\end{aligned} \tag{11}$$

, where $H(\Psi^*(t), \Psi(t-1))$ is a shortend notation for the hamiltonian to remind that the two different flavours of Grassmann field enters in the expression of the Hamiltonian with different time-index. However, in the time-continuum limit when we would take, $N \rightarrow \infty \mid \epsilon \rightarrow 0$, we shall lose this distinction for the Hamiltonian. We substitute the expression for the matrix element in the original expression for the partition

function and find that in the continuum limit,

$$\begin{aligned}
Z &= \lim_{\epsilon \rightarrow 0} \int_{\psi(N)=-\psi(1)} \prod_{t=1}^{t=N-1} \left(\prod_i d\Psi_i^*(t) d\Psi_i(t) \right) e^{-\sum_{t=2}^N \epsilon \left[\sum_i \left(\frac{\Psi_i^*(t) \Psi_i(t) - \Psi_i^*(t) \Psi_i(t-1)}{\epsilon} \right) + H(\Psi^*(t), \Psi(t-1)) \right]} \\
&= \int_{\psi(\beta)=-\psi(0)} \mathcal{D}[\Psi^*, \Psi] e^{-\int_0^\beta d\tau (\sum_{ij} \Psi_i^*(t) [\delta_{ij} (\partial/\partial\tau - \mu) + t_{ij}] \Psi_j(t) + \sum_{ijkl} U_{ijkl} \Psi_i^*(t) \Psi_j^*(t) \Psi_k(t) \Psi_l(t))}
\end{aligned} \tag{12}$$

, where the shortened expression, $\mathcal{D}[\Psi^*, \Psi] = \prod_{t=1}^{t=N-1} (\prod_i d\Psi_i^*(t) d\Psi_i(t))$ has been introduced. The sense of this expression is understood as time ordered integral of all possible functional over all possible paths. Algebraically, any transformation for the Grassmann fields would also transform this integral measure by a factor of the Jacobian of the transformation. The quantity in the exponent of the above expression is known as the Euclidean action or imaginary time action, $S_E(\Psi^*, \Psi) = \int d\tau [\Psi^* (\partial/\partial\tau) \Psi + H(\Psi^*, \Psi)]$ (with suppressed indices).

C. Hubbard-Stratanovich Transformation

The reformulation of the Fermionic problem in the language of functional integrals, or as it is more commonly referred in the literature, the path integral, makes way for utilising various controlled approaches towards solving the problem. If the model that we are discussing for the moment did not have the four-fermion interaction term, it would have been readily soluble by diagonalising the hopping matrix t in real space. If the underlying lattice of the model has translational symmetry then the dimensionality of the diagonalisation problem could be reduced by fourier transforming the Grassmann variable. E.g. for a one-dimensional lattice without any additional sublattice degrees of freedom would transform the non-interacting Euclidean action into a simple diagonal form for the fourier modes, $\Psi(t)_k$. The differential operator can also be tackled by doing a temporal fourier transform, $\Psi(t)_k = \frac{1}{\sqrt{\beta}} \sum_{\omega_n} \Psi_k(n) e^{-i\omega_n \tau}$. In this way the partion function can be reduced to multiple of Gaussian integrals and can be exactly evaluated and can be found to agree with results obtained from single particle quantum mechanical calculations.

However, the real benefit of the functional integral formalism emerges in the presence of many-body interactions when standard quantum-mechanical calculations break down. In the presence of weak interaction the exponential in the functional integral can be formally expanded in a Taylor series of interaction strength U and pertarbative results can be obtained. In principle the calculation in such a scheme would boil down to computing multiple orders of Gaussian integrals with the 'free' two-body part of the action appearing as the quadratic exponent of the Gaussian integral. As a look from the expression of the partition function would reveal, this is far from being a straightforward exercise as multiple mode of the Grassmann

fields can combine in different ways in the expansion to give non-zero contribution. Wick's theorem (to be discussed shortly) gives a formal way to collect these contributions. Feynman's diagram makes the job of applying Wick's theorem easier in a pictorial way. All interactions would not be weak and there are other non-perturbative methods like Wilsonian RG, where the functional integral is carefully integrated out for the higher energy modes and thereby resulting into a functional integral of fields only involving the lower energy modes. In this procedure often the action changes dynamically and then assumes a simple ultimate form. A lot of interesting discovery of condensed matter physics in the past century came through this systematic way of looking at a physical model in different energy scales. There are still other methods involving numerical computation of the functional integral using approximate techniques like the Monte-Carlo sampling.

For this section we shall look at a particular method of recasting the interacting Fermionic functional integral problem to a coupled but non-interacting Bosonic-Fermionic problem. This method, named as the Hubbard-Stratanovich transformation, remains an exact description of the original problem. However, it leads way to novel approximation schemes (as we shall see later) and can also be used to look at effective field theories at different energy scales. The transformation utilises basic identities of Grassmann and scalar Gaussian integrals which we list ahead of the formal H-S transformation of our model problem.